ECMA 31150 – Spring 2026

Midterm Exam

April 21, 2026

Instructor: Mikkel Plagborg-Moller

- You have 75 minutes to complete the following 5 problems.
- The test is closed-book. You may consult a calculator and one double-sided sheet of handwritten notes.
- The total number of points is 100.
- The problems have not been ordered in terms of difficulty. Many of the questions can be answered independently of each other.
- **Show your work**. State several significant digits for every numerical answer.
- **Partial answers will receive partial credit.** Do not leave questions empty. However, points may be subtracted for long, irrelevant remarks.
- Good luck!

Problem 1 (10 points total)

Let the time series $\{y_t\}$ be defined by

$$y_t \equiv (u_t + u_{t-1})u_{t-1} \quad \text{for } t = 1, 2, \ldots, \quad \text{where} \quad u_t \overset{iid}{\sim} N(0,1) \quad \text{for } t = 0, 1, 2, \ldots$$

i) (10 points.) Show that $\{y_t\}$ is covariance stationary. Derive its mean, its ACF, and its long-run variance. [Hint: the 4th moment of the standard normal distribution equals 3.]

Problem 2 (20 points total)

Let $\{y_t\}$ be a covariance stationary process with autocovariance function $\gamma(\ell) \equiv \text{Cov}(y_t, y_{t-\ell})$. Let $\hat{\sigma}_y^2 \equiv \frac{1}{T-1}\sum_{t=1}^{T}(y_t - \bar{y})^2$ denote its sample variance, where $\bar{y} \equiv \frac{1}{T}\sum_{t=1}^{T} y_t$.

i) (12 points.) Prove that

$$E\left[\hat{\sigma}_y^2\right] = \gamma(0) - \frac{2}{T-1}\sum_{\ell=1}^{T-1}\left(1 - \frac{\ell}{T}\right)\gamma(\ell).$$

[Hint: it is without loss of generality to assume $E[y_t] = 0$.]

ii) (8 points.) Give a concrete example of a covariance stationary model for $\{y_t\}$ such that $\hat{\sigma}_y^2$ is a strictly *downward-biased* estimator of the population variance $\gamma(0)$.

Problem 3 (20 points total)

Consider the AR(2) process

$$y_t = -2 - \tfrac{1}{4}y_{t-1} + \tfrac{1}{8}y_{t-2} + u_t,$$

where $\{u_t\}$ is a white noise process with variance 2, and $E[u_t \mid \{y_s\}_{s<t}] = 0$.

i) (8 points.) Verify that the process is covariance stationary, provided the initial values have appropriate moments. [Note: you do not need to derive the required moments of the initial values.]

ii) (12 points.) Give an explicit formula for the optimal two-step-ahead forecast $y_{T+2|T} \equiv E[y_{T+2} \mid \{y_s\}_{s\leq T}]$? What is the associated RMSFE?

Problem 4 (26 points total)

We have $T = 196$ observations of two time series $\{y_t, x_t\}$. We assume an ADL model for y_t with two autoregressive lags and one distributed lag of x_t. A regression produces the following output:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.469
Model:                            OLS   Adj. R-squared:                  0.460
Method:                 Least Squares   F-statistic:                     49.42
Date:                Tue, 21 Apr 2026   Prob (F-statistic):           1.18e-23
Time:                        11:00:00   Log-Likelihood:                -274.71
No. Observations:                 194   AIC:                             557.4
Df Residuals:                     190   BIC:                             570.5
Df Model:                           3
Covariance Type:                  HC1
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.0101      0.072     -0.140      0.889      -0.152       0.132
y_lag1         0.6422      0.070      9.172      0.000       0.505       0.779
y_lag2        -0.0719      0.062     -1.156      0.247      -0.194       0.050
x_lag1         0.1971      0.053      3.728      0.000       0.093       0.301
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
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i) (6 points.) Suppose the last observations in the data set are $y_T = 1$, $x_T = 0$, $y_{T-1} = -1$, $x_{T-1} = 3$. What is the optimal forecast of y_{T+1}?

ii) (6 points.) Propose a null hypothesis for the model coefficients under which $\{y_t\}$ is white noise. Then test this hypothesis, using a 5% significance level. [Note: if you're unsure how to use the regression output to answer this question, you will receive full points if you provide a detailed general description of how to conduct the test.]

iii) (6 points.) Test whether $\{x_t\}$ Granger-causes $\{y_t\}$, using a 10% significance level.

iv) (8 points.) Assume that $\{y_t, x_t\}$ are jointly covariance stationary. Treating the above coefficient estimates as exact rather than estimated, derive an expression for $E[y_t]$ as a function of $E[x_t]$.

Problem 5 (24 points total)

Indicate whether each of the following statements is true or false. Justify your answer with a very brief argument.

i) (8 points.) Suppose we observe a sample of size T from the time series $\{\tilde{y}_t\}$ given by $\tilde{y}_t \equiv (1 - \frac{1}{2}L)^{-1} y_t$, where $\{y_t\}$ follows a stationary AR(1) process with non-zero AR coefficient. Then the BIC procedure will select 2 lags with arbitrarily high probability as $T \to \infty$.

ii) (8 points.) A bivariate VAR(1) process with coefficient matrix

$$\boldsymbol{A} = \begin{pmatrix} 0.8 & 0 \\ 1.5 & -0.3 \end{pmatrix}$$

is covariance stationary (provided the initial value has appropriate moments).

iii) (8 points.) If we run the recursive residual bootstrap for a VAR(1) model, using a very large number of iterations, then the sample average of the bootstrap draws $\hat{A}^*_{ij}$ will be very close to the real-data parameter estimate $\hat{A}_{ij}$. [Note: an informal argument suffices for this question.]